Group Theory

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Outline

1. What is a Group?
2. Abelian Groups
3. Subgroup
4. Cyclic Group
5. Cosets
6. Quotients

What is a Group?

Definition of a Group

Definition: A **Group** is some set G, along with a binary operation $\circ$ that conforms some **Axioms**.

Notation: We write a Group as $(G, \circ)$ to explicitly specify both the set G and the operation $\circ$, or simply G with any operation.

Example

The set G can be any set as long as it is non-empty
Ex:

- Hours in a clock form a set $H = \{0, 1, 2, ..., 11\}$
- Integer set: $\mathbb{Z} = \{..., -2, -1, 0, 1, 2, ...\}$
- $\mathbb{R}$, $\mathbb{N}$, $\mathbb{Q}$, $\mathbb{C}$
- Even numbers $2\mathbb{Z} = \{..., -2, 0, 2, ...\}$

Example

The binary operation $\circ$ can be any function as long as it takes 2 inputs and produces 1 output, e.g $f(a, b) = c$ (from which the name "binary")
Ex:

- Arithmetic: Addition $(+)$, Multiplication $(\times)$,...
- Boolean Algebra: XOR $(\oplus)$, AND $(\wedge)$, OR $(\vee)$
- Others: $a \circ b = a^b$, $a \circ b = a \times b + 2a - 3$

Definition of a Group

This system must satisfy the following **4 Axioms**:

1. **Closure**: For all $a, b \in G$, the result $a \circ b$ is also in G.
2. **Associativity**: For all $a, b, c \in G$:

$$(a \circ b) \circ c = a \circ (b \circ c)$$

3. **Identity Element**: There exists an element $e \in G$ such that:

$$a \circ e = a = e \circ a$$

4. **Inverse Element**: For every $a \in G$, there exists $a^{-1} \in G$ such that:

$$a \circ a^{-1} = e = a^{-1} \circ a$$

Example

Let's test if Integers $\mathbb{Z} = \{\ldots, -1, 0, 1, \ldots\}$ under Addition $(+)$ follow the rules.

Example

- **1. Closure:** Does sum of 2 integers still stays inside the set?
  - *Check:* $5 + (-3) = 2$. Is 2 an integer? **Yes.**
  - You can't add two whole numbers and suddenly get a fraction (like 2.5).

Example

- **1. Closure:** Does sum of 2 integers still stays inside the set?
  - *Check:* $5 + (-3) = 2$. Is 2 an integer? **Yes.**
  - You can't add two whole numbers and suddenly get a fraction (like 2.5).
- **2. Associativity:** Does the order of addition matter?
  - *Check:*

$$(2 + 3) + 4 = 5 + 4 = \mathbf{9}$$

$$2 + (3 + 4) = 2 + 7 = \mathbf{9}$$

  - The sums are identical.

Example

- **3. Identity Element:** Is there an Identity element?
  - We need an e such that $a + e = a$.
  - *Check 0:* $5 + 0 = 5$ and $(-7) + 0 = -7$.
  - **Yes**, $e = 0$ is the identity.

Example

- **3. Identity Element:** Is there an Identity element?
  - We need an e such that $a + e = a$.
  - *Check 0:* $5 + 0 = 5$ and $(-7) + 0 = -7$.
  - **Yes**, $e = 0$ is the identity.
- **4. Inverse Element:** Can we "undo" an addition?
  - For any integer a, can we get back to the identity (0)?
  - If we have 5, we need -5 to reach 0.
  - If we have -100, we need 100 to reach 0.
  - **Yes**, for every a, the inverse is $-a$.

Example

So $(\mathbb{Z}, +)$ is indeed a **group**, what about $(\mathbb{N} = \{0, 1, 2, \dots\}, +)$?

Example

Let's check it out:

- Closure, Associativity, Identity: Similar to $(\mathbb{Z}, +)$
- Inverse: Is there inverses for every $a \in N$?
  Unfortunately no, there's no number in N that satisfies: $1 + x = 0$

Therefore, $(\mathbb{N}, +)$ is not a **group**.

Example

We proved $(\mathbb{Z}, +)$ is a group and $(\mathbb{N}, +)$ isn't (infinite set).
What about finite sets?

Example

Consider the set Integer modulo n $\mathbb{Z}_n$ under addition operator.
Let's test if it is a group or not:

- Closure, Associativity, Identity: Similar to $(\mathbb{Z}, +)$
- Inverse:
  - For any $a \in \mathbb{Z}_n$, we want $b \in \mathbb{Z}_n$ such that

$$a + b \equiv 0 \pmod{n}$$

  - Choose $b = n - a$.
  - Then:

$$a + (n - a) = n \equiv 0 \pmod{n}$$

  - So every element has an inverse.

Therefore, $(\mathbb{Z}_n, +)$ is a group.

Properties of a Group

Uniqueness: The Identity element (e) and Inverses (a^{-1}) are unique.
Cancellation Law: For all $a, b, c \in G$:

$$\text{If } a \circ b = a \circ c, \text{ then } b = c$$

Solvability: The equation $a \circ x = b$ always has a unique solution $x \in G$

We'll prove the first property and leave the rest in Exercises

Properties of a Group

Proposition 1: The identity element of a group is unique.

Proof:
Suppose e and e' are both identity elements in G.

$$\begin{aligned} e \circ e' &= e' \quad \text{(treating } e \text{ as the identity)} \\ e \circ e' &= e \quad \text{(treating } e' \text{ as the identity)} \end{aligned}$$

Combine the two results:

$$e' = e \circ e' = e$$

Therefore, $e = e'$, so the identity is unique. $\square$

Properties of a Group

Proposition 2: The inverses in a group are unique.

Proof:
Suppose g' and g'' are both inverse of element g in G. We need to show that $g' = g''$.
We have:

$$g \circ g' = e = g' \circ g$$
$$g \circ g'' = e = g'' \circ g$$

$\rightarrow g' = g' \circ e = g' \circ (g \circ g'') = (g' \circ g) \circ g'' = e \circ g'' = g''$
Therefore, $g' = g''$, so the every inverse for all $g \in G$ is unique. $\square$

Exercises

Exercises

Exercise 1: Show that for all $a, b, c \in G$:

$$\text{If } a \circ b = a \circ c, \text{ then } b = c$$

Exercise 2: Show that the equation $a \circ x = b$ always has a unique solution $x \in G$

Abelian Groups

Abelian Groups

Definition: A group $(G, \circ)$ is called **Abelian** (or commutative) if it satisfies one extra axiom:

5. Commutativity: For all $a, b \in G$:

$$a \circ b = b \circ a$$

Example

Abelian:

- Integers $(\mathbb{Z}, +)$ because $1 + 2 = 2 + 1$. The same applies to $(\mathbb{R}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{C}, +)$
- Multiplicative Group modulo p $(\mathbb{Z}_p, \times)$, because multiplication is commutative

Non-Abelian:

- Matrix Multiplication, because usually $A \times B \neq B \times A$.

Exercises

Exercises

Exercise 1: Let $G = \{x \in R : x > 1\}$ and $a \circ b = a \times b - a - b + 2$ for all $a, b \in R$. Show that:

a) G is closed under that operation.

b) The set G under the operation $\circ$ forms an abelian group

Subgroups

What is Subgroup?

Similarly to a set, subgroup is a "special" version of subset
Recall that a set H' is a subset of H if and only if $H' \subseteq H$, in which H contain all elements of H'

So what is more special about a subgroup?

What is Subgroup?

Definition: A **Subgroup** is a subset $H \subseteq G$ that forms a group in its own right **under the same binary operation** as G.

Or in simpler term, a group within a group.

We write $H \leq G$ as H is a **subgroup** of G.

Properties of Subgroups

The Subgroup Test: Instead of checking all 4 axioms, we only need to verify two conditions for a non-empty subset H:

1. **Closure:** $\forall a, b \in H \implies a \circ b \in H$
2. **Inverses:** $\forall a \in H \implies a^{-1} \in H$

Note: Properties like Associativity and Commutativity are automatically inherited from the parent group G.

Example

Consider the set $2\mathbb{Z} = \{\ldots, -4, -2, 0, 2, 4, \ldots\}$.

This set forms a group under addition, denoted as $(2\mathbb{Z}, +)$.

Relationship: Since every even integer is also an integer ($2\mathbb{Z} \subseteq \mathbb{Z}$), we say that:

$$(2\mathbb{Z}, +) \text{ is a } \textbf{subgroup} \text{ of } (\mathbb{Z}, +)$$

Example

Similiar to previous example, we also got:

$$H = \{0, 3\} \text{ and } K = \{0\} \text{ are subgroups of } \mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$$

Under addition operator.

Notice that the subset $K = \{e_G\}$ is always a subgroup of G.
We call this group is a **trivial subgroup** while a proper subset H is called a **proper subgroup**.

Properties of Subgroups

Identity: The identity of G is in H.
Not a Proposition: This is not a proposition but you should know that the subset $H \subseteq G$ could be a **group** without being a **subgroup** of G.

Proof

Proposition 1: The identity of G (e_G) is in H (e_H).
It is quite easy to verify that the identity of G is in H.

Assume that $a \in H$ where $a \neq e_G$.
Since H is a **group**, a^{-1} must be in H and so is $a \circ a^{-1} = e_H$.
Looking from the group G perspective, $a \circ a^{-1}$ is also equal to e_G.

Therefore, $e_H = e_G \in H$.

Cyclic Group

Cyclic Group

Alright, knowing all elements in a group G is tough, especially in a large group.

So...

Is there any ways we can "know" or "generate" all elements just from a single element alone?

Cyclic Group

For example:

The Sequence $(\mathbb{Z}, +)$:

$$\ldots \xrightarrow{+1} \underbrace{-2(1)}_{-2} \xrightarrow{+1} \underbrace{-1(1)}_{-1} \xrightarrow{+1} \underbrace{0(1)}_{0} \xrightarrow{+1} \underbrace{1(1)}_{1} \xrightarrow{+1} \underbrace{2(1)}_{2} \xrightarrow{+1} \ldots$$

Cyclic Group

Let G be an Abelian group and $a \in G$. We denote the set

$$\langle a \rangle := \{a^k : k \in \mathbb{Z}\} = \{\dots, a^{-1}, e, a, a^2, \dots\} \subseteq G \quad \text{(Multiplicative notation)}$$

$$\langle a \rangle := \{ka : k \in \mathbb{Z}\} = \{\dots, -a, e, a, 2a, \dots\} \subseteq G \quad \text{(Additive notation)}$$

For example, in $(\mathbb{Z}, +)$:

$$\langle 1 \rangle = \{\dots, -1, 0, 1, \dots\} = \mathbb{Z}$$

$$\langle 3 \rangle = \{\dots, -3, 0, 3, \dots\} = 3\mathbb{Z}$$

$$\langle 7 \rangle = \{\dots, -7, 0, 7, \dots\} = 7\mathbb{Z}$$

Cyclic Group

For $a \in G$, we call $\langle a \rangle$ the **cyclic subgroup** generated by a.

Definition: A group G is **cyclic** if there exists an element $a \in G$ such that $G = \langle a \rangle$. Such an element a is called a **generator** of G.

Example

Let take a look at $(\mathbb{Z}_6, +)$.
If we try to evaluate $\langle a \rangle = \{e, a, 2a, \ldots, 5a\} \in \mathbb{Z}_6$ of this group:

Element a	**1a**	**2a**	**3a**	**4a**	**5a**	**6a**	**7a**	**Generated Subgroup** $\langle a \rangle$
0	0	0	0	0	0	0	0	$\{0\}$
1	1	2	3	4	5	0	1	$\mathbb{Z}_6$
2	2	4	0	2	4	0	2	$\{0, 2, 4\}$
3	3	0	3	0	3	0	3	$\{0, 3\}$
4	4	2	0	4	2	0	4	$\{0, 2, 4\}$
5	5	4	3	2	1	0	5	$\mathbb{Z}_6$

Cyclic Group

Let G be a group and $a \in G$.

- We define the **order** of a, denoted by $ord(a)$, to be the smallest positive integer k such that $a^k = e$.
- If no such integer exists, we say $ord(a) = \infty$.

Key Properties:

1. $a^i = a^j$ if and only if $i \equiv j \pmod{ord(a)}$.
2. The cyclic subgroup $\langle a \rangle$ has exactly $ord(a)$ elements.
3. (Lagrange) In a finite group G, $ord(a)$ always divides $|G|$. (We'll prove this later hehe)

Example

Recall our group $(\mathbb{Z}_6, +)$.

Element a	**1a**	**2a**	**3a**	**4a**	**5a**	**6a**	**Generated Subgroup** $\langle a \rangle$
0	0	0	0	0	0	0	$\{0\}$
1	1	2	3	4	5	0	$\mathbb{Z}_6$
2	2	4	0	2	4	0	$\{0, 2, 4\}$
3	3	0	3	0	3	0	$\{0, 3\}$
4	4	2	0	4	2	0	$\{0, 2, 4\}$
5	5	4	3	2	1	0	$\mathbb{Z}_6$

The order of every elements are $\{1, 6, 3, 2, 3, 6\}$ respectively. And all of them are divide $|G| = 6$.

Cyclic Group

In a finite group, we define the **exponent** of a finite group G, denoted $exp(G)$, is the smallest positive integer m such that:

$$a^m = e \quad \text{for all } a \in G$$

Observations:

- $m = exp(G) = \text{lcm}\{ord(a) \mid a \in G\}$.
- By Lagrange's Theorem, $exp(G)$ always divides $|G|$.

If G is a finite **abelian** group, there *always* exists an element $a \in G$ such that $ord(a) = exp(G)$.

Cyclic Group

Example: Take a look at $G = \mathbb{Z}_{15}^*$.

$$G = \{1, 2, 4, 7, 8, 11, 13, 14\}$$

a	**Powers (mod15)**	**Order** $ord(a)$
1	$1^1 = 1$	1
2	$2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16 \equiv 1$	4
4	$4^1 = 4, 4^2 = 16 \equiv 1$	2
7	$7^1 = 7, 7^2 = 49 \equiv 4, 7^3 = 28 \equiv 13, 7^4 \equiv 1$	4
8	$8^1 = 8, 8^2 = 64 \equiv 4, 8^3 = 32 \equiv 2, 8^4 \equiv 1$	4
11	$11^1 = 11, 11^2 = 121 \equiv 1$	2
13	$13^1 = 13, 13^2 = 169 \equiv 4, 13^3 = 52 \equiv 7, 13^4 \equiv 1$	4
14	$14^1 = 14, 14^2 = 196 \equiv 1$	2

In which, the exponent of the group is LCM(1,2,4) $= 4$ while G's order is 8

Cyclic Group

A group G is **cyclic** if $G = \langle a \rangle$ for some $a \in G$. We distinguishes two cases:

Case 1: Infinite Cyclic Group If $ord(a) = \infty$, then all powers a^z are distinct.

$$\langle a \rangle = \{\dots, a^{-2}, a^{-1}, e, a, a^1, a^2, \dots\}$$

Example: $(\mathbb{Z}, +)$ with generator 1.

Case 2: Finite Cyclic Group If $ord(a) = n < \infty$, then n is the **smallest positive integer** such that $a^n = e$.

$$\langle a \rangle = \{e, a, a^2, \dots, a^{n-1}\}$$

Example: $(\mathbb{Z}_n, +)$ with generator $[1]$.

Cyclic Group

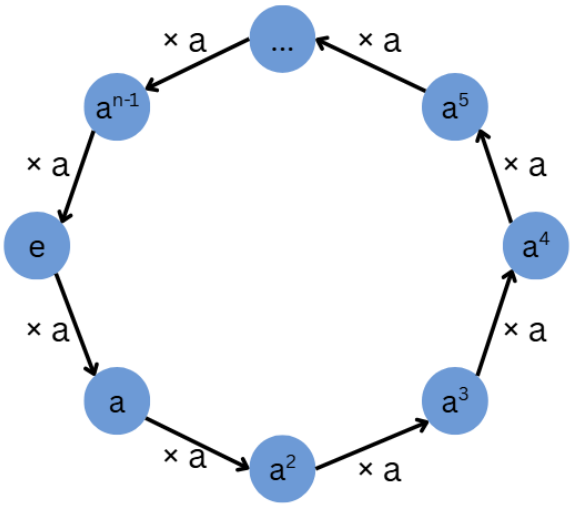

Figure: Why this

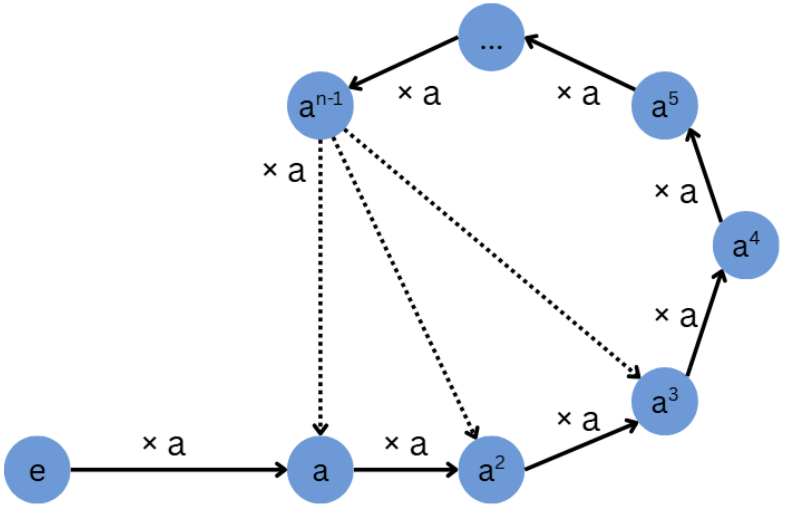

Figure: But not this?

Proof

Let $\langle a \rangle = \{e, a, a^2, \ldots, a^{n-1}\}$ has n **distinct** elements.
Assume that $a^n = a^x$ for any $0 < x < n$.

Since every element has inverse

$$a^n a^{-x} = a^x a^{-x} \to a^{n-x} = e$$

So e appears **twice** in $\langle a \rangle$, result in breaking the rule n distinct elemts.

Therefore a^n has to be e.

Properties of Cyclic Groups

Let $G = \langle a \rangle$ be a cyclic group, then:

Every cyclic group is abelian

Proof: Any $x, y \in G$ can be written as $x = a^m,\ y = a^n$ for some $m, n \in \mathbb{Z}$. Then

$$xy = a^m a^n = a^{m+n} = a^{n+m} = a^n a^m = yx.$$

So G is abelian.

Every subgroup of a cyclic group is cyclic

Proof idea: Let $H \leq G$. If $H = \{e\}$ it is cyclic. Otherwise, pick the smallest $k > 0$ such that $a^k \in H$. Then every element of H is a power of a^k, so $H = \langle a^k \rangle$.

Cosets

Cosets

Let H be a subgroup of G. For any element $a \in G$, we can "shift" H by applying the group operation to a and every element of H.

Definition:
The **left coset** of H in G represented by a is the set:

$$aH = \{a \circ h \mid h \in H\}$$

Right coset can be defined similarly by:

$$Ha = \{h \circ a \mid h \in H\}$$

Example

Let examine the group $(G, +)$ with $G = \mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$.
Consider the subset $H = \{0, 3\}$.

Brief proof that H is a subgroup:

- Closure: $0 + 0 = 0$, $0 + 3 = 3$, $3 + 3 = 6 \equiv 0 \pmod 6$ (All in H).
- Inverses: Since the identity element of H is 0
  Inverses of 0 is 0 and 3 is 3.

The cosets of H are:

$$0 + H = 3 + H = \{0, 3\}$$
$$1 + H = 4 + H = \{1, 4\}$$
$$2 + H = 5 + H = \{2, 5\}$$

Properties of Cosets

Identity: For any $g \in G$, $gG = G$

Partition: The set of all cosets partitions the group G. This means every element of G belongs to exactly one left coset.

Equal Size: All cosets have the same cardinality as the subgroup H: $|aH| = |Ha| = |H|$.

Lagrange: Let G be a finite group and let H be a subgroup of G. Then $|G|/|H| = [G : H]$ (index of a subgroup H in a group G) is the number of distinct left cosets of H in G.

Properties of Cosets

Proposition 1: For any $g \in G$, $gG = G$.

Let $G = \{g_1, g_2, \ldots, g_n\}$. We want to show that $gG = \{g \circ g_1, g \circ g_2, \ldots, g \circ g_n\}$ is the same set as G.

By the **Closure Property** of groups, for any $g_i \in G$, the product $g \circ g_i$ must also be an element of G.

Thus, every element in gG is in G ($gG \subseteq G$).

Properties of Cosets

Suppose $g \circ g_i = g \circ g_j$.
By the **Cancellation Law**, we have $g_i = g_j$. This means all n elements in gG are distinct.

Since gG has n distinct elements and $gG \subseteq G$, it must be that $gG = G$.

Properties of Cosets

Proposition 2: The set of all cosets partitions the group G.
Let g_1H and g_2H be two cosets of H in G. We must show that either $g_1H \cap g_2H = \emptyset$ or $g_1H = g_2H$.

Suppose that $g_1H \cap g_2H \neq \emptyset$ and $a \in g_1H \cap g_2H$.

Then by the definition of a left coset, $a = g_1 \circ h_1 = g_2 \circ h_2$ for some elements h_1 and h_2 in H. Hence, $g_1 = g_2h_2h_1^{-1}$.

Properties of Cosets

Substitute $g_1 = g_2 h_2 h_1^{-1}$ into $g_1 H$:

$$g_1 H = (g_2 h_2 h_1^{-1}) H$$

Since $h_2 h_1^{-1} \in H$, then $h_2 h_1^{-1} H = H \rightarrow \ \mathrm{g}_1 H = g_2 H$.

Therefore, cosets never partially overlap. Since every element g is in at least one coset ($g \in gH$), the group G is perfectly partitioned.

Properties of Cosets

Proposition 3: All cosets have the same cardinality as the subgroup H: $|aH| = |Ha| = |H|$.
Similarly to first one, we want to show that $H = \{h_1, h_2, h_3, \dots\}$ and $aH = \{a \circ h_1, a \circ h_2, a \circ h_3, \dots\}$ are essentially the same.

By using **Cancellation Law**, we prove that $|H| = |aH|$.
The same goes to Ha, therefore, $|aH| = |Ha| = |H|$.

Properties of Cosets

Proposition 4: Let G be a finite group and let H be a subgroup of G. Then $|G|/|H| = [G : H]$ is the number of distinct left cosets of H in G.

The group G is partitioned into $[G : H]$ distinct left cosets. Each left coset has $|H|$ elements.

Therefore, $|G| = [G : H]|H|$.

Quotients

Normal Group

A subgroup H of a group G is **normal** in G if $g \circ H = H \circ g$ for all $g \in G$.

That is, a **normal** subgroup of a group G is one in which the right and left cosets are precisely the same.

Example

Let G be an abelian group. Every subgroup H of G is a **normal** subgroup.

Since $g \circ h = h \circ g$ for all $g \in G$ and $h \in H$, it will always be the case that $gH = Hg$.

Quotient

We denote $G/H = \{g_1H, g_2H, ...\}$ is a set of all cosets of subgroup H in G.
Ex: $\mathbb{Z}/2\mathbb{Z} = \{0 + 2\mathbb{Z}, 1 + 2\mathbb{Z}\}$.

Quotients

Let N is a **normal** subgroup of a group G.

Then the list of all cosets of N in G form a group G/N:

$$G/N = \{g_1N, g_2N, g_3N, g_4N, ...\}$$

Under the operation $A \circ B = (aN) \circ (bN) = abN$ for all $A, B \in G/N$.

This group is called the **factor** or **quotient** group of G and N.

Question: You might wonder why $(aN) \circ (bN) = abN$? Let slow down a bit.

Example

Let A and B be **Cosets** of the normal subgroup N in G under **addition** operator.
Then A and B can be interpreted as:

$$A = aN = a + N = a + \{n_1, n_2, ...\} = \{n_1 + a, n_2 + a, ...\}$$
$$B = bN = b + N = b + \{n_1, n_2, ...\} = \{n_1 + b, n_2 + b, ...\}$$

$\rightarrow (aN) + (bN) = \{n_i + n_j + a + b\}$ for $i, j \in [1, 2, \ldots, |N|]$

Example

Since N must be closure and there exists an identity element in N, we can reduce the $n_i + n_j$ to:

$$(aN) + (bN) = \{n_i + a + b\} \text{ for } i \in [1, 2, \ldots, |N|]$$
$$\rightarrow (aN) + (bN) = (a + b) + N = abN$$

Since $(a + b) \in G$, hence $abN \subseteq G/N \ \forall \ a, b \in G$. Therefore the operation is closed.

Let continue on proving G/N is a group.

Proof

We know the operation is closed. Now for the other three axioms. G/N inherits its structure from G.

1. Associativity:
Let $A = aN$, $B = bN$, $C = cN$. Since $(ab)c = a(bc)$ in G:

$$\begin{aligned}(A \circ B) \circ C &= (abN) \circ cN = (ab)cN \\ &= a(bc)N = aN \circ (bcN) = A \circ (B \circ C)\end{aligned}$$

2. Identity Element:
The identity in G/N is the coset $eN = N$.

$$(aN) \circ (eN) = (ae)N = aN$$

$$(eN) \circ (aN) = (ea)N = aN$$

Proof

3. Inverse Elements:
For any coset aN, its inverse is the coset $a^{-1}N$.
When we combine them, we should get the identity element (N):

$$(aN) \circ (a^{-1}N) = (aa^{-1})N = eN = N$$
$$(a^{-1}N) \circ (aN) = (a^{-1}a)N = eN = N$$

Since G/N satisfies closure, associativity, identity, and invertibility, the quotient group G/N is indeed a valid group.

Example

Consider the normal subgroup $3\mathbb{Z}$ of $\mathbb{Z}$ under addition. The cosets of $3\mathbb{Z}$ in $\mathbb{Z}$ are:

$$0 + 3\mathbb{Z} = \{\dots, -3, 0, 3, 6, \dots\}$$
$$1 + 3\mathbb{Z} = \{\dots, -2, 1, 4, 7, \dots\}$$
$$2 + 3\mathbb{Z} = \{\dots, -1, 2, 5, 8, \dots\}$$

Then the group $\mathbb{Z}/3\mathbb{Z}$ is given by the Caley table below:

$+$	$0+3\mathbb{Z}$	$1+3\mathbb{Z}$	$2+3\mathbb{Z}$
$0+3\mathbb{Z}$	$0+3\mathbb{Z}$	$1+3\mathbb{Z}$	$2+3\mathbb{Z}$
$1+3\mathbb{Z}$	$1+3\mathbb{Z}$	$2+3\mathbb{Z}$	$0+3\mathbb{Z}$
$2+3\mathbb{Z}$	$2+3\mathbb{Z}$	$0+3\mathbb{Z}$	$1+3\mathbb{Z}$

Properties

Abelian Inheritance: If G is an Abelian group, then G/N is also an Abelian group.

Cyclic Inheritance: If G is a cyclic group, then G/N is also a cyclic group.

Proof

Proposition 1: If G is Abelian, then G/N is Abelian.

Let aN and bN be any two arbitrary elements in the quotient group G/N. By the definition of coset multiplication in a quotient group, we have:

$$(aN)(bN) = (ab)N$$

Since G is an Abelian group, its elements commute, meaning $ab = ba$. Substituting this back into our coset equation:

$$(ab)N = (ba)N = (bN)(aN)$$

Since $(aN)(bN) = (bN)(aN)$ for all elements, G/N is Abelian. $\square$

Proof

Proposition 2: If G is cyclic, then G/N is cyclic.

Let $G = \langle g \rangle$, meaning g is a generator of G. We want to show that gN generates G/N. Let xN be an arbitrary element in G/N. Since $x \in G$ and

G is generated by g, there exists some integer k such that $x = g^k$. Therefore, substituting x into our coset:

$$xN = (g^k)N = (gN)^k$$

Since every element xN in G/N can be written as a power of gN, the element gN is a generator. Thus, $G/N = \langle gN \rangle$ and is cyclic. $\square$

This suffering has came to an end. Thank you for listening